

1.

```
xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ bankers
===== BANKER'S ALGORITHM =====

Initial system safety check:
System is in a SAFE state.
Safe sequence: P1 -> P3 -> P4 -> P0 -> P2

--- Scenario 1: Safe Request ---

P1 requests: [1 0 2]
Request granted.
Safe sequence: P1 -> P3 -> P4 -> P0 -> P2

--- Scenario 2: Unsafe Request ---

P4 requests: [0 3 0]
Request denied - would lead to unsafe state
$ |
```

2.

```
xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ deadlockdetect
===== DEADLOCK DETECTION USING WAIT-FOR GRAPH =====

===== SCENARIO 1: NO DEADLOCK =====
Wait-for graph:
P0 -> P1
P1 -> P2
P2 -> P3
No deadlock detected.

===== SCENARIO 2: DEADLOCK =====
Wait-for graph:
P0 -> P1
P1 -> P2
P2 -> P0
Deadlock cycle: P0 -> P1 -> P2 -> P0
Deadlock detected.
$
```

3.

```
xv6 kernel is booting

init: starting sh
$ resourceorder
=====
  RESOURCE ORDERING DEADLOCK DEMO
=====

=====
  BAD RESOURCE ORDERING
=====

Process A: Lock1 -> Lock2
Process B: Lock2 -> Lock1

DEADLOCK CREATED:
A holds Lock1 and waits for Lock2
B holds Lock2 and waits for Lock1
Circular wait: A -> B -> A
$
```

```
init: starting sh
$ resourceorder
=====
RESOURCE ORDERING DEADLOCK DEMO
=====

=====
FIXED RESOURCE ORDERING
=====
Both processes use: Lock1 -> Lock2
Process A: started
Process A: requesting Lock1
Process A: acquired Lock1
Process A: requesting Lock2
Process A: acquired Lock2
Process A: doing work
Process A: releasing Lock2
Process A: releasing Lock1
Process A: completed
Process B: started
Process B: requesting Lock1
Process B: acquired Lock1
Process B: requesting Lock2
Process B: acquired Lock2
Process B: doing work
Process B: releasing Lock2
Process B: releasing Lock1
Process B: completed
Fixed ordering test completed successfully.
$ |
```

4.

```
xv6 kernel is booting

init: starting sh
$ syncdeadlock
=====
  COMBINED SYNCHRONIZATION & DEADLOCK
  AVOIDANCE DEMONSTRATION
=====

Resource instances:
Printer = 2
Scanner = 1
Disk    = 2

Deadlock prevention strategy:
RESOURCE ORDERING
Global order: Printer < Scanner < Disk

P0: ===== Cycle 1 =====
P0: requesting Printer
P0: granted Printer
P0: requesting Scanner
P0: granted Scanner
P0: starting work
P0: releasin
P1: ===== Cycle 1 =====
P1: requesting Printer
P1: granted Printer
P1: requesting Disk
P1: granted Disk
P1: starting work
```

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P1: releasing Disk
P1: releasing Printer
P1: cycle completed

P1: ===== Cycle 2 =====
P1: requesting Printer
P1: granted Printer
P1: requesting Disk
P1: granted Di
P2: ===== Cycle 1 =====
P2: requesting Scanner

P3: ===== Cycle 1 =====
P3: requesting Printer

P4: ===== Cycle 1 =====
P4: requesting Printer
g Scanner
P0: releasing Printer
P0: cycle completed

P0: ===== Cycle 2 =====
P0: requesting Printer
P0: granted Printer
P0: requesting Scanner
P0: granted Scanner
P0: starting work
P0: releasing Scanner
P0: releasing Printer
P0: cycle completed
```

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P0: ===== Cycle 3 =====
P0: requesting Printer
P0: granted Printer
P0: requesting Scanner
P0: granted Scanner
P0: starting work
P0: releasing Scanner
P0: releasing Printer
P0: cycle completed
sk
P1: starting work
P1: releasing Disk
P1: releasing Printer
P1: cycle completed

P1: ===== Cycle 3 =====
P1: requesting Printer
P1: granted Printer
P1: requesting Disk
P1: granted Disk
P1: starting work
P1: releasing Disk
P1: releasing Printer
P1: cycle completed

P1: ALL CYCLES COMPLETED
P2: granted Scanner
P2: requesting Disk
P2: granted Disk
P2: starting work
P2: releasing Disk
P2: releasing Scanner
```

P2: cycle completed

P2: ===== Cycle 2 =====

P2: requesting Scanner

P2: granted Scanner

P2: requesting Disk

P2: granted Disk

P2: starting work

P3: granted Printer

P3: requesting Scanner

P4: granted Printer

P4: requesting Disk

P4: granted Disk

P4: starting work

P4: releasing Disk

P4: releasing Printer

P4: cycle completed

P4: ===== Cycle 2 =====

P4: requesting Printer

P4: granted Printer

P4: requesting Disk

P4: granted Disk

P4: starting work

P4: releasing Disk

P4: releasing Printer

P4: cycle completed

P4: =

P0: ALL CYCLES COMPLETED

P2: releasing Disk

```
P2: releasing Disk
P2: releasing Scanner
P2: cycle completed

P2: ===== Cycle 3 =====
P2: requesting Scanner
P2: granted Scanner
P2: requesting Disk
P2: granted Disk
P2: starting work
P2: releasing Disk
P2: releasing Scanner
P2: cycle completed

P2: ALL CYCLES COMPLETED
P3: granted Scanner
P3: s===== Cycle 3 =====
P4: requesting Printer
P4: granted Printer
P4: requesting Disk
P4: granted Disk
P4: starting work
P4: releasing Disk
P4: releasing Printer
P4: cycle completed

P4: ALL CYCLES COMPLETED
tarting work
P3: releasing Scanner
P3: releasing Printer
P3: cycle completed
```

```
P3: ===== Cycle 2 =====  
P3: requesting Printer  
P3: granted Printer  
P3: requesting Scanner  
P3: granted Scanner  
P3: starting work  
P3: releasing Scanner  
P3: releasing Printer  
P3: cycle completed
```

```
P3: ===== Cycle 3 =====  
P3: requesting Printer  
P3: granted Printer  
P3: requesting Scanner  
P3: granted Scanner  
P3: starting work  
P3: releasing Scanner  
P3: releasing Printer  
P3: cycle completed
```

```
P3: ALL CYCLES COMPLETED
```

```
=====  
ALL 5 PROCESSES COMPLETED  
NO DEADLOCK OCCURRED  
=====  
$ |
```